



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 1 Examinations 2023/2024

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2BA1, 2BCM1, 2BCT1, 2BMU1, 2BDA1, 2BGE1, 2BGG1,
2BGS1, 2BME1, 2BS1, 2CSM1, 2EH1, 2EHS1,
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3BS9, 3MR3, 1CSD1, 1DVV1

Exams First Science, Second Arts, Second Science, Third Science,
Masters in Computer Science and
Higher Diploma in Data Analytics and Visualisation

Module Code **ST2001**

Module **Statistics for Data Science I**

School School of Mathematical and Statistical Sciences

External Examiner Prof. N. Bargary
Internal Examiners *Dr Nicola Fitz-Simon
Prof. C. Scarrott

Instructions: **Answer all questions**

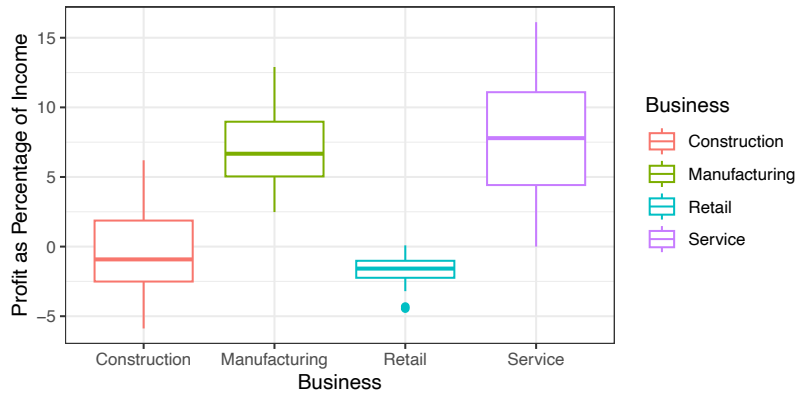
Duration 2 Hours
No. of Pages 18 pages, including this one

Requirements:

Release to Library:	No: ☒	Yes: ☐	
Release in Exam Venue	No: ☒	Yes: ☐	
Handout	No: ☒	Yes: ☐	
Formulae & Tables	No: ☒	Yes: ☐	Tables and Formulae attached
Cambridge Tables	No: ☒	Yes: ☐	
Graph paper	No: ☒	Yes: ☐	
Other Materials	No: ☐	Yes: ☒	Any non-communicating calculator allowed
Colour Graphics	No: ☒	Yes: ☐	

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1. Data on the profit (as a percentage of income) for 25 businesses of each of four different business types (construction, retail, manufacturing or service) are summarised in the box-plots below.



Choose one of the following options that best describe the two variables **Business** and **Profit**, respectively.

- (a) (Quantitative, Quantitative)
 - (b) (Quantitative, Qualitative)
 - (c) (Qualitative, Qualitative)
 - (d) (Qualitative, Quantitative)
2. A health expert believes that the “average salt consumed daily by Irish children under 10 years old is 1.5 teaspoons.”

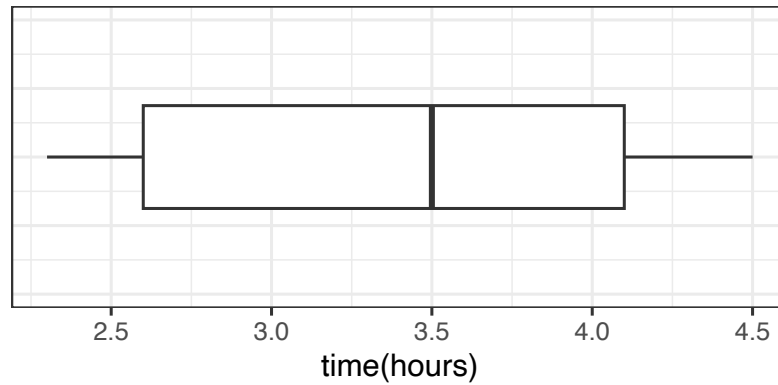
A simple random sample of size 100 from the population of such children is surveyed and asked to keep a food diary to monitor salt consumption. The mean from those surveyed is 1.45 teaspoons.

Which ONE of the following statements is **INCORRECT**:

- (a) The value 1.5 teaspoons is a parameter and the value 100 is a statistic.
- (b) The value 1.5 teaspoons is a parameter and the value 1.45 teaspoons is a statistic.
- (c) The value 1.5 teaspoons is a statistic and the value 100 is a parameter.
- (d) The value 100 is a statistic and the value 1.45 teaspoons is a statistic.

3. The duration of charge of a simple random sample of mobile phones is measured, and the resulting durations in hours are (2.6, 3.2, 4.1, 2.3, 3.6, 2.4, 4.3, 3.5, 4.5).

A boxplot of these data is shown below:

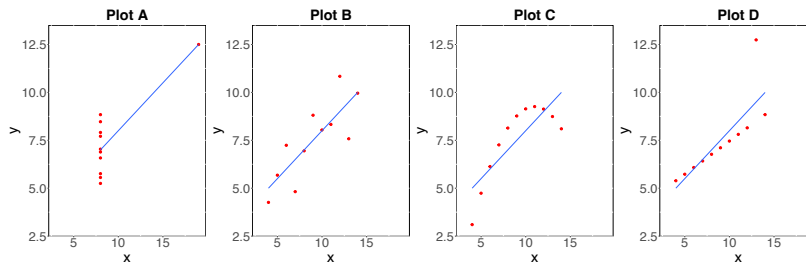


What is the correct interpretation of the distribution of the duration of charge in the population of mobile phones from the boxplot?

- (a) The distribution is unimodal and symmetric
 - (b) The distribution has negative skew
 - (c) The distribution is bimodal
 - (d) The distribution has positive skew
 - (e) The sample is too small to draw conclusions about the distribution using a boxplot
4. A sample of six students studying Science were asked how many hours they spent travelling to College in a typical week. Their responses were:
6, 9, 9, 8, 9, 9
Find the sample median of these data.
(Note: input the answer to 1 decimal place.)
5. Which of the following measures of spread is **most sensitive** to outliers?
- (a) Variance
 - (b) Standard deviation
 - (c) Range
 - (d) Interquartile range

6. Each of the following four plots are scatterplots of 11 datapoints. For each scatterplot the correlation between x and y is the same, correlation = 0.82. The least squares line is also the same for each scatterplot, with slope = 0.5 and intercept = 3.

For which of these plots is linear regression appropriate?



- (a) Plot C only
 (b) Plot A only
 (c) Plot B only
 (d) All the plots
 (e) None of the plots
 (f) Plot D only
7. Which one of the following statements best describes the difference between an observational study and an experiment?
- (a) In an experiment the researcher is careful to determine exactly what is to be observed, whereas in an observational study there is greater concern about how the observations are made.
 (b) In an experiment treatments are deliberately imposed upon individuals and their responses are measured. In an observational study there is no such intervention, individuals are observed and variables of interest are measured.
 (c) In an experiment, blocking is always used as this will improve the precision of the estimates, whereas improving the precision of estimates is not possible in an observational study.
 (d) Without concrete examples, it is not possible to detect what the differences might be.
8. A university wishes to know the satisfaction levels of undergraduate students. They want to estimate satisfaction of both male and female students in each discipline. Some disciplines have a relatively large number of males compared to females, or vice versa. What sampling scheme is the most efficient to get a sample that is representative of all groups of interest?
- (a) Random sample at the level of cluster, with clusters defined by discipline and gender
 (b) Simple random sampling
 (c) Actively recruit students who are willing to express their views
 (d) Stratify by discipline and gender and take a random sample within each stratum
9. How many distinct ways can the letters in the word SCIENTIFIC be arranged?
10. A rowing coach has a squad of 14 rowers and wants to choose 8 of them to row in a race. How many different choices can the coach make, if order does not matter?

11. Among pirates, 38% own a talking parrot, 27% have a wooden leg, and 14% both own a talking parrot and have a wooden leg.

What is the probability that a randomly selected pirate does not own a parrot and does not have a wooden leg?

(Note: input the answer to 2 decimal places.)

12. If $P(A) = 0.13$, $P(B) = 0.16$, and $P(A \cup B) = 0.28$. Select which one of the following describes the two events A and B.

- (a) A and B are independent
- (b) A and B are neither mutually exclusive nor independent
- (c) A and B are both mutually exclusive and independent
- (d) A and B are mutually exclusive

13. Pig A goes to market with probability 0.25 if pig B goes to market, and pig B goes to market with probability 0.50 if pig A goes. Sadly, at least one pig must go to market - that is, if we let A be the event that pig A goes to market, and B be the event that pig B goes to market, then $P(A \cup B) = 1$.

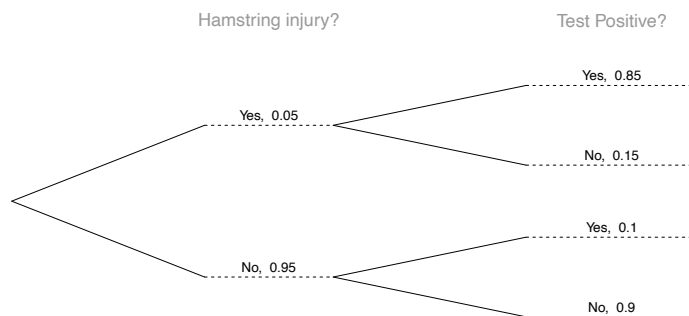
What is the probability that both pigs go to market? **select one answer:**

- (a) $3/4$
- (b) $1/20$
- (c) 1
- (d) 0
- (e) $1/5$
- (f) It is not possible to calculate this probability without further information

14. The proportion of football players who have a hamstring injury any week during the season is 5%.

A new test to alert players to a potential hamstring injury before it occurs detects 85% of hamstring injuries. The test also incorrectly identifies 10% of fit players as being at risk of a hamstring injury.

These probabilities are represented in the following tree diagram.



What is the probability that a randomly selected player will get a hamstring injury alert?

(Note: input the answer to 3 decimal places.)

15. It is thought that 5% of athletes take a specific performance-enhancing drug.

A drug test is able to detect the drug 85% of the time when it is present. The test also detects the drug 5% of the time when it is not present.

These probabilities are shown in the following tree diagram. At the end of each branch is the probability of following that branch.



Which of the following is the probability that a randomly selected athlete who has a positive drug test is taking the drug?

- (a) 0.05
 (b) 0.0425
 (c) 0.85
 (d) 0.472
16. The returns, in euro, of a gambling game is given by a random variable X , where X has the following probability distribution:

x	-2	-1	0	1	2
P(X = x)	0.2	0.4	0.1	0.2	0.1

Find the expected return, $E(X)$.

(Note: input the answer to 1 decimal place.)

17. Three seeds are selected at random from a large batch of seeds and planted. It is known that the number of seeds that germinate is given by a random variable X , where X has the following probability distribution:

x	0	1	2	3
P(X = x)	0.15	0.25	0.3	0.3

The expected number of seeds that germinate is $E(X) = 1.75$. Find the standard deviation of the number of seeds that germinate, $sd(X)$.

(Note: round the answer to 2 decimal places where relevant.)

18. In which one of the following examples could X be modelled by a Binomial random variable?
- (a) A group of six friends go to Mr Waffle, X is the number of the group that order a hot chocolate
 - (b) 20 water samples are taken from a river at different distances downstream from a factory, X is the number of samples that test positive for the presence of a pollutant produced by the factory
 - (c) Calls arrive at a call centre as a random process, X is the number of calls arriving between 9am and 5pm
 - (d) Ten medical devices are selected as a simple random sample from a production line, X is the number of defective devices in the sample
19. It is reported that eating dark chocolate is beneficial to the heart. Suppose that 55% of people in a population prefer milk chocolate to dark chocolate. 12 people were randomly selected from this population and asked whether they prefer milk or dark chocolate. Use the binomial distribution to calculate the probability that exactly 4 of them prefer milk chocolate?
(Note: round the answer to 3 decimal places where relevant.)
20. Assume goals scored by both teams in normal playing time in a football match occur as a Poisson process, with a mean of 2.8. Calculate the probability that more than one goal is scored in a randomly selected match.
(Note: round the answer to 3 decimal places where relevant.)
21. If a random variable X , follows a Poisson distribution with $\lambda = 14$, i.e. $X \sim \text{Poisson}(14)$. Which of the following normal distributions would be a good approximation of the distribution of X ?
- (a) $X \sim N(14, 14)$
 - (b) $X \sim N(14, 1)$
 - (c) $X \sim N(0, 14)$
 - (d) $X \sim N(0, 1)$
22. A household recycling bin collection service states that the weight collected each fortnight follows a normal distribution with mean 21.75 kg and standard deviation 9.05 kg. What proportion of customer's recycling bins weigh between 3.66 kg and 39.84 kg?
(Note: input the answer to 3 decimal places. Use the standard normal tables.)
23. A household waste bin collection service states that the weight collected each fortnight follows a normal distribution with mean 16.05 kg and standard deviation 5.26 kg. The amount of waste in the lightest 97.5% of the bins is below what value?
(Note: input the answer to 3 decimal places. Use the standard normal tables.)
24. Suppose that a simple random sample of 81 bags of potatoes are weighed in grams, and a 95% confidence interval calculated for the population mean weight has limits (976.08, 983.92). Given this sample size is large enough to use the Central Limit Theorem (CLT), calculate the standard error in the estimate of the population mean weight.
(Note: round the answer to 3 decimal places where relevant.)

25. The Department of Education wishes to estimate the proportion of full-time third level students who are in paid employment for more than 8 hours per week during term time. They have planned to recruit a simple random sample of third level students. They want the half-width of the 95% confidence interval (margin of error) to be 8%. How many students should they recruit? (Hint: assume the 'worst case' value for the proportion, $p = 1/2$.)

(Note: your answer should be rounded up to the nearest integer.)

26. A simple random sample of Irish adults is surveyed about gambling habits, and the sample percentage of "problem gamblers" is 12.1%. The sample is also used to estimate that we are 95% confident that the population percentage of problem gamblers is between 10.1% and 14.1%.

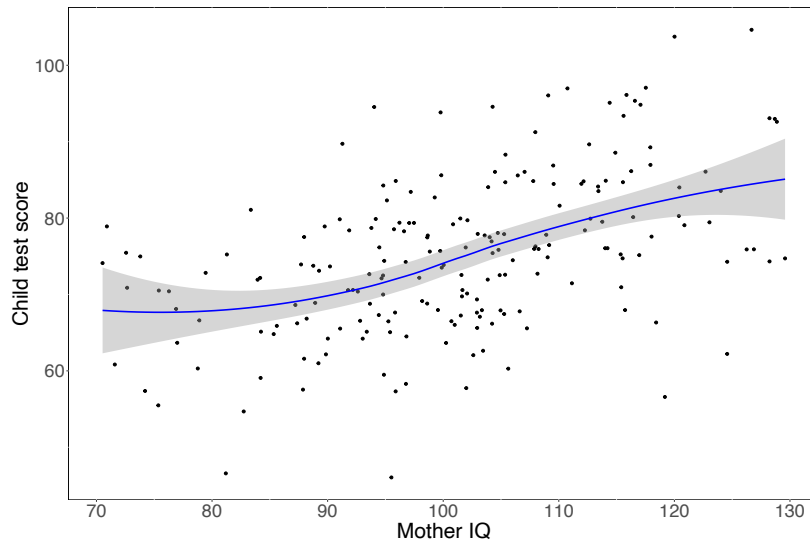
Which **ONE** of the following interpretations of this confidence interval is **CORRECT**?

- (a) There is a 95% chance that the sample problem gambling percentage is contained in the interval
 - (b) For 95% of people in the population, the problem gambling percentage is between 10.1% and 14.1%
 - (c) There is a 95% probability that the problem gambling percentage in the population is between 10.1% and 14.1%
 - (d) Because we are not 100% confident, the interval does not give us any information about the population proportion of problem gambling
 - (e) There is a 5% chance that the sample problem gambling percentage is not contained in the interval
 - (f) It is quite likely that the problem gambling percentage in the population is contained in the interval
27. Which one of the following would result in a confidence interval estimate that is more precise, i.e. narrower?
- (a) If the standard error was increased
 - (b) If the mean of the sample was equal to the population mean
 - (c) Decreasing the significance level
 - (d) Decreasing the confidence level
 - (e) Taking a sample that is smaller in size
 - (f) If the population was more variable, i.e. larger standard deviation

28. Aliens have abducted a simple random sample of 8 medical students from the University of Galway. All they know about medical students is that their heights follow a normal distribution. Based on previous abductions of other groups, they are interested in whether the true mean height of medical students at the University of Galway is different to 160cm, but acknowledge the difference could be in either direction. What are their null and alternative hypotheses?
- (a) H_0 : true mean height equals 0cm; H_1 : true mean height is greater than 0cm
 - (b) H_0 : sample mean height does not equal 160cm; H_1 : sample mean height equals 160cm
 - (c) H_0 : true mean height equals 160cm; H_1 : sample mean height is greater than 160cm
 - (d) H_0 : sample mean height equals 160cm; H_1 : sample mean height does not equal 160cm
 - (e) H_0 : true mean height equals 160cm; H_1 : true mean height does not equal 160cm
 - (f) H_0 : true mean height does not equal 160cm; H_1 : true mean height equals 160cm
29. The aliens calculate the mean height for their sample is 169cm, and sample standard deviation is 5cm (recall the sample size is 8).
Based on these sample statistics, calculate an appropriate test statistic for the aliens' **hypothesis in the previous question**.
What is the **smallest** value for the **upper bound of the p -value** that can be determined from the t -table?
30. Based on the **two previous questions**, what is the appropriate statistical test the aliens should use for inference and why?
- (a) The assumptions needed to use a normal or a t -test are not met, the only way the aliens can make inferences is by bootstrapping
 - (b) As it is a simple random sample and the observations are normal, CLT applies, so the aliens should use a normal test
 - (c) As it is a simple random sample CLT applies, so they can use a normal test
 - (d) As it is a small simple random sample the aliens can use a t -test; the normality of the observations is not relevant
 - (e) It is a small simple random sample and the observations are normally distributed so the aliens can use a t -test

31. Researchers interested in the relationship between mother's IQ and child's test score at age 8 collect a random sample of 200 mother-child pairs. Mothers were asked to complete an IQ test and the result recorded as *ma_iq* (X). The child's school test result at age 8 was recorded as *child_test* (Y).

A scatterplot (with a **smoother**) of the data is as follows:



The estimated correlation coefficient is 0.49. Use this result to determine the **strength, direction and form of relationship** between mother IQ and child test score.

- (a) (weak/moderate, positive, non-linear)
- (b) (strong, negative, non-linear)
- (c) (strong, positive, linear)
- (d) (weak/moderate, negative, linear)
- (e) (weak/moderate, positive, linear)
- (f) (strong, positive, non-linear)
- (g) (strong, negative, linear)
- (h) (weak/moderate, negative, non-linear)

32. The researchers continue to study the relationship between child's test score at age 8 and maternal IQ described in the **previous question**. For a random sample of 200 mother-child pairs, consider the model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ (for $i = 1, \dots, 50$) under the usual regression assumptions.

The output below provides the fit of a regression analysis carried out on these data.

```
# Simple linear regression model
lm(child_test ~ ma_iq, data = df)

##
## Call:
## lm(formula = child_test ~ ma_iq, data = df)
##
## Coefficients:
## (Intercept)      ma_iq
##      37.470      0.378
```

Using the output provided above indicate which one of the following statements is **TRUE**:

- (a) The regression coefficient indicates that for each additional point of mother's IQ the child's test score decreases by 0.378 on average.
 - (b) The line of best fit is $\text{mean}(\text{ma_iq}) = 37.4701 + 0.378 * \text{child_test}$
 - (c) The estimated intercept, indicates that for each additional point of mother's IQ the child's test score increases by 37.4701 on average.
 - (d) The regression coefficient indicates that for each additional point of mother's IQ the child's test score increases by 0.378 on average.
 - (e) The line of best fit is $\text{mean}(\text{ma_iq}) = 0.378 + 37.4701 * \text{child_test}$
 - (f) The line of best fit is $\text{mean}(\text{child_test}) = 0.378 + 37.4701 * \text{ma_iq}$
33. Consider the data and linear regression model in the **two previous questions**. Would you be concerned about using the linear regression model to predict the test score for a child whose mother had an IQ score of 150, and why?
- (a) No, as the linear model was fit in a random sample of mother-child pairs
 - (b) Yes, as the coefficient of determination is low
 - (c) Yes, as this is extrapolating beyond the range of observed data
 - (d) No, it is reasonable to assume the linear model provides an accurate prediction

ST2001 Formula Sheet

Descriptive Statistics

Sample mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Sample standard deviation

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Counting theory

Permutations

$${}_n P_r = \frac{n!}{(n-r)!}$$

Combinations

$${}_n C_r = \binom{n}{r} = \frac{n!}{(n-r)!}$$

Probability

For an events A and B in a sample space S :

$$0 \leq P(A) \leq 1$$

$$P(S) = 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional probability

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Multiplication rule

$$P(A \cap B) = P(B|A)P(A)$$

Independence of A and B

$$P(B|A) = P(B) \text{ or } P(A|B) = P(A) \text{ or } P(A \cap B) = P(A)P(B)$$

Bayes Rule

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Total Probability: if $E_1, E_2, \dots, E_k$ are a set of mutually exclusive and exhaustive events:

$$P(A) = P(A \cap E_1) + P(A \cap E_2) + \dots + P(A \cap E_k) = P(A|E_1)P(E_1) + \dots + P(A|E_k)P(E_k)$$

Random Variables

A random variable X is a function mapping the outcomes of a random experiment to a numeric value in $\mathbb{R}$.

Discrete random variable X has observed values x and a probability mass function $p(x)$ with

1. $p(x) \geq 0$
2. $\sum_x p(x) = 1$
3. $P(X = x) = p(x)$

Mean

$$\mu = E(X) = \sum_x xP(X = x)$$

Variance

$$\sigma^2 = Var(X) = \sum (x - \mu)^2 P(X = x) = E(X^2) - E(X)^2$$

Continuous random variable X has observed values x and a probability density function $f(x)$ with

1. $f(x) \geq 0$
2. $\int_x p(x)dx = 1$
3. $\int_a^b dx = P(a < X < b)$

$P(X = x) = 0$ for all x for a continuous random variable.

Distributions

Name	Representation	Range	pmf/pdf	$E(X)$	$Var(X)$
Binomial	$X \sim Binomial(n, p)$	$x \in \{0, 1, 2, \dots, n\}$	$P(X = x) = \binom{n}{x} x^p (n - x)^{1-p}$	np	$np(1 - p)$
Poisson	$X \sim Poisson(\lambda)$	$x \in \{0, 1, 2, \dots\}$	$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$	λ	λ
Normal	$X \sim N(\mu, \sigma^2)$	$x \in \mathbb{R}$	$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$	μ	σ^2

Confidence intervals

Large sample $100(1 - \alpha)\%$ confidence intervals for population:

$$\begin{array}{ll} \text{mean } \mu & \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \\ \text{proportion } \pi & p \pm z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} \end{array}$$

Small sample $100(1 - \alpha)\%$ confidence intervals for population:

$$\text{mean } \mu \quad \bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

where $t_{\alpha/2}$ is on $n - 1$ degrees of freedom.

Note: $\bar{x}$ is the sample mean, s is the sample standard deviation, p is the sample proportion, n is the sample size.

Sample size calculations

For margin of error no greater than e when estimating:

$$\begin{array}{ll} \text{mean } \mu & n = \left(\frac{z_{\alpha/2} \sigma}{e} \right)^2 \\ \text{proportion } \pi & n = \frac{(z_{\alpha/2})^2 \pi(1-\pi)}{e^2} \end{array}$$

Hypothesis tests based on a single sample

Large sample test statistic for a population:

$$\text{mean } \mu \quad z = \frac{\bar{x} - \mu_0}{\sigma_{\bar{x}}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \quad \text{where } H_0 : \mu = \mu_0$$

Small sample test statistic for a population:

$$\text{mean } \mu \quad t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad \text{where } H_0 : \mu = \mu_0$$

and t is on $n - 1$ degrees of freedom.

Note: $\bar{x}$ is the sample mean, s is the sample standard deviation, n is the sample size.

Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X + \epsilon_i, i = 1, \dots, n$$

β_0 is the intercept, β_1 is the slope.

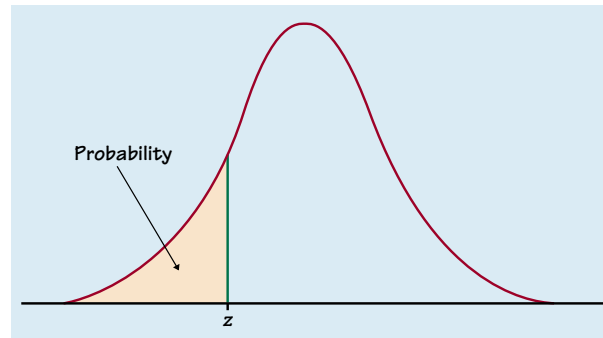


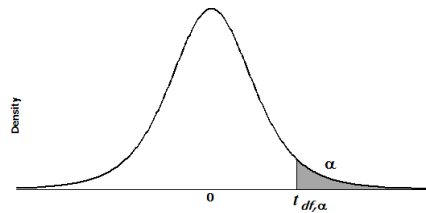
Table entry for z is the probability lying below z .

TABLE A Standard normal probabilities

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

[illegible]

Table: t distribution critical values



df	Upper tail probability											
	0.25	0.20	0.15	0.10	0.05	0.025	0.02	0.01	0.005	0.0025	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.706	15.895	31.821	63.657	127.321	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.089	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
Z*	0.674	0.842	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.291